

# Where and When Young Drivers Crash: Targeting Motor Premium Loadings Across Post-COVID Victoria

A business report for RACV motor underwriting

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## 1 Executive Summary

This report investigates how crash severity and young-driver involvement vary across time of day and Local Government Areas (LGAs) in post-COVID Victoria using Victorian Road Crash Data from 2018 to 2024. Results indicate that crash frequency increases substantially

during afternoon and early evening periods, while several LGAs, including Hume, Casey, and Melton, recorded comparatively higher rates of crashes involving drivers aged 18–25. Although geographic variation was moderate, findings suggest that both time-of-day and location provide useful contextual indicators of crash risk. RACV may benefit from incorporating temporal and geographic risk considerations alongside driver age when assessing motor insurance premium loading decisions.

## 2 Introduction

Road crashes remain a significant public safety and financial concern across Victoria, contributing to injury, loss of life, and substantial insurance costs. Young drivers are often considered a higher-risk group due to limited driving experience and greater exposure to risky driving behaviours. Following the COVID-19 pandemic, traffic patterns and travel behaviours shifted considerably, potentially changing the spatial and temporal distribution of crash risk across the state. Understanding where and when crashes involving young drivers are more likely to occur can provide valuable insight for risk assessment and insurance decision-making.

This report investigates how crash severity and young-driver involvement vary by time of day and Local Government Area (LGA) across post-COVID Victoria using Victorian Road Crash Data from 2018 to 2024. The analysis is framed from the perspective of an RACV motor underwriting analyst seeking to identify geographic and temporal risk factors relevant to premium loading decisions. Particular attention is given to young-driver involvement, crash severity, and regional variation to identify areas of elevated risk. Findings from this analysis may assist insurers in developing more targeted and evidence-based underwriting strategies.

## 3 Methodology and Data Preparation

Table 1: Summary of key variables used in the analysis.

| Variable     | Role_in_analysis                                                                        |
|--------------|-----------------------------------------------------------------------------------------|
| SEVERITY     | Crash outcome category used to compare fatal, serious injury, and other injury crashes. |
| hour         | Derived from accident time and used to analyse time-of-day crash patterns.              |
| LGA_NAME     | Used to compare geographic variation across Local Government Areas.                     |
| DTP_REGION   | Used for broad regional comparison across Victoria.                                     |
| young_driver | Binary indicator showing whether a crash involved at least one driver aged 18–25.       |

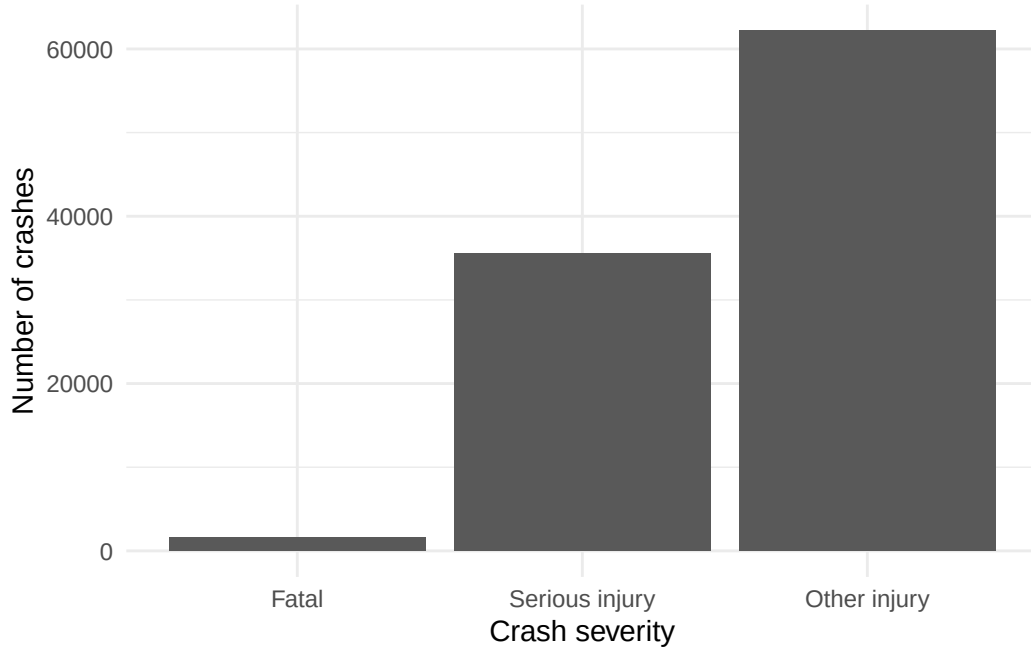


Figure 1: Crash severity composition in the cleaned Victorian road crash dataset, 2018–2024.

Victorian Road Crash Data was obtained from the Victorian Department of Transport and Planning (Victorian Department of Transport and Planning (2026)) and filtered to include crashes occurring between 1 January 2018 and 31 December 2024. The dataset contains information on crash severity, location, road conditions, vehicle involvement, and demographic indicators relevant to motor insurance risk assessment. Data preparation involved removing non-injury accident records and excluding entries associated with the discontinued “Metropolitan South East Region” category to improve consistency across regional comparisons. Placeholder values, such as unknown speed zones, were recoded as missing values, while additional analytical variables were constructed, including crash hour and a binary indicator of young-driver involvement, defined as crashes involving at least one driver aged 18–25. Crash severity was also recoded into an ordered factor comprising Fatal, Serious injury, and Other injury categories.

As shown in Figure 1, the cleaned dataset was first examined according to crash severity composition to understand the crash outcomes included in the analysis. Table Table 1 summarises the key variables used to investigate temporal and geographic crash patterns. The Results section then examines time-of-day crash patterns and Local Government Area variation in young-driver crash involvement to support RACV motor underwriting decisions.

## 4 Results

### 4.1 Temporal Crash Patterns

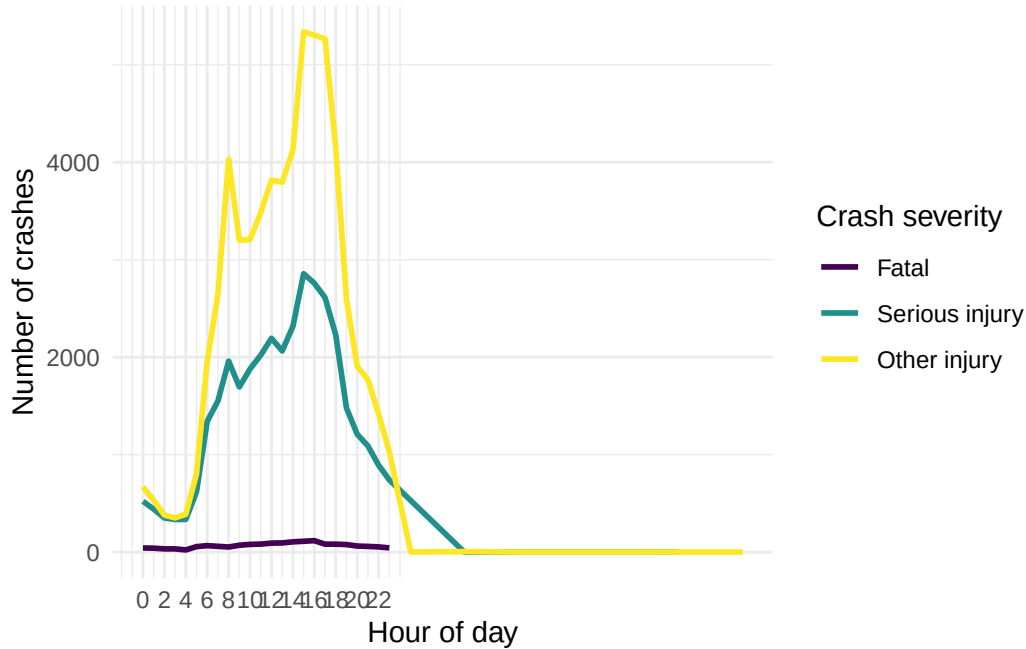
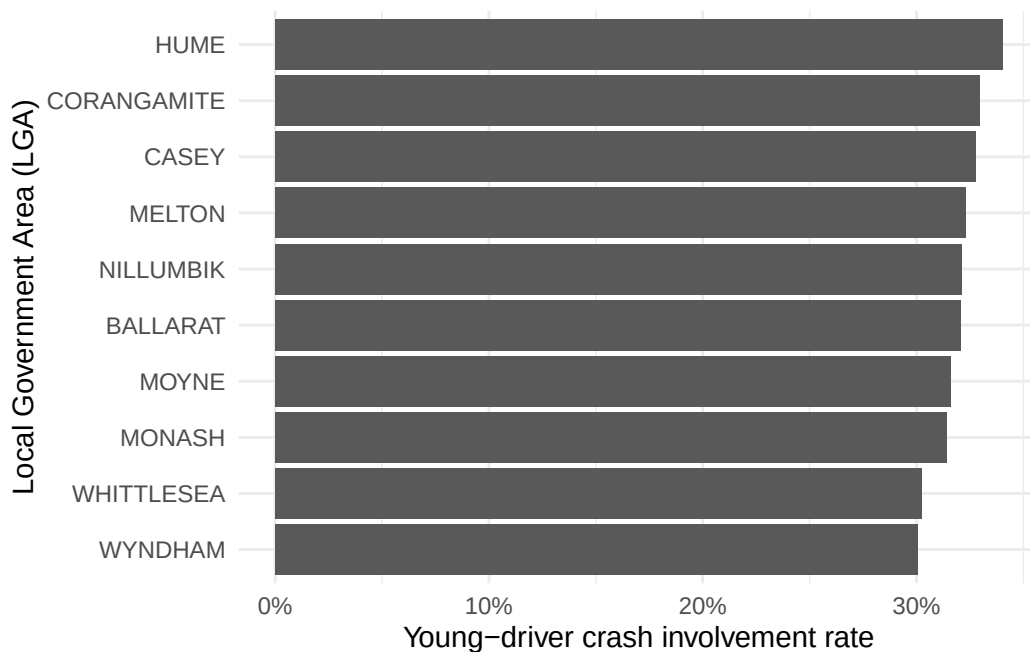


Figure 2: Crash frequency by hour of day and severity in Victoria, 2018–2024.

As shown in @fig-temporal-patterns, crash frequency varied considerably across the day between 2018 and 2024. Crashes involving all severity levels generally increased during morning commuting hours, followed by a more pronounced rise during the afternoon and early evening period. The highest concentration of crashes occurred between approximately 3:00 pm and 5:00 pm, while crash frequency was substantially lower during early morning hours between 3:00 am and 5:00 am.

Other injury crashes consistently represented the largest category across all hours, while serious injury and fatal crashes remained comparatively less frequent. However, serious and fatal crashes still followed a similar temporal pattern, with increased activity during peak travel periods. These findings suggest that time of day may be an important contextual factor for underwriting decisions, particularly during periods of higher traffic exposure when crash frequency increases substantially.

## 4.2 Geographic results for young-driver crash analysis



As shown in @fig-young-driver-lga, young-driver crash involvement varied across Victorian Local Government Areas (LGAs) between 2018 and 2024. Hume recorded the highest proportion of crashes involving drivers aged 18–25, followed by Corangamite, Casey, and Melton, with young-driver involvement rates exceeding 30% of total crashes. Several outer metropolitan LGAs, including Casey, Whittlesea, and Wyndham, also appeared among the highest-ranked areas, indicating geographic variation in crashes involving young drivers. However, differences between LGAs were relatively modest, suggesting that elevated young-driver involvement occurs across multiple regions rather than being isolated to a small number of areas. From an underwriting perspective, these findings indicate that geographic context may provide useful supporting information for premium loading decisions when considered alongside driver age and temporal crash risk indicators.

## 5 Discussion

The findings indicate that both temporal and geographic factors are relevant when considering crash risk involving young drivers in post-COVID Victoria. Temporal analysis showed that crash frequency increased substantially during afternoon and early evening periods, particularly during likely commuting hours, suggesting that time of day may influence overall crash exposure. Although fatal crashes remained comparatively uncommon, serious and other injury crashes

followed similar temporal patterns, indicating that elevated crash activity is not limited to lower-severity incidents.

Geographic analysis identified moderate variation in young-driver crash involvement across Local Government Areas (LGAs). Hume, Casey, Melton, and several outer metropolitan or regional LGAs recorded comparatively higher proportions of crashes involving drivers aged 18–25. While differences between LGAs were not extreme, the findings suggest that crash involvement among young drivers is not evenly distributed across Victoria and may vary according to local traffic conditions, demographic composition, or travel patterns.

Several limitations should be acknowledged. The dataset does not include measures of traffic exposure, kilometres travelled, or behavioural factors, meaning causal explanations cannot be established. In addition, post-COVID travel behaviour may continue to evolve over time, which may influence future crash trends.

## 6 Conclusion

This report examined how crash severity and young-driver involvement varied across time of day and Local Government Areas in post-COVID Victoria using Victorian Road Crash Data from 2018 to 2024. Results suggest that crash frequency is concentrated during afternoon and early evening periods, while several LGAs recorded comparatively higher proportions of crashes involving drivers aged 18–25. Together, these findings indicate that both temporal and geographic context may provide useful information when assessing motor insurance risk.

## 7 Recommendations

Based on the findings, RACV may benefit from incorporating time-of-day and geographic risk indicators alongside driver age when assessing premium loading decisions. Rather than relying solely on age-based assumptions, insurers may achieve more targeted and evidence-based underwriting by considering contextual factors associated with elevated crash involvement. Future analysis may also benefit from incorporating traffic exposure measures or behavioural indicators to improve risk assessment accuracy.

## References

Victorian Department of Transport and Planning. 2026. “Victoria Road Crash Data.” <https://opendata.transport.vic.gov.au/dataset/victoria-road-crash-data>.